Convex optimization for machine learning – TD4

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1 Exercises

Exercise 1. (Projected stochastic gradient)
We consider the following optimization problem

$$\min_{\mathbf{x}\in C} \sum_{i=1}^{n} f_i(\mathbf{x}) \tag{1}$$

where $C = [0,1]^d$ and for all i, $f_i : \mathbb{R}^d \to \mathbb{R}$ is differentiable.

1. We define the convex indicator function of the set C as $\iota_C(x) = \begin{cases} 0 & \text{if } \mathbf{x} \in C \\ +\infty & \text{if } \mathbf{x} \notin C \end{cases}$.
   Show that (1) is equivalent to $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{i=1}^{n} f_i(\mathbf{x}) + \iota_C(\mathbf{x})$.
2. Compute the proximal operator of ι_C
3. Write the proximal stochastic gradient method for the resolution of (1).

Exercise 2. (Optimisation with explicit constraints)
We consider the following optimization problem

$$\min_{\mathbf{x}\in C} f(\mathbf{x}) \tag{2}$$

where $C \subset \mathbb{R}^d$ is a convex set and $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and convex.

1. We define the convex indicator function of the set C as
   $$\iota_C(\mathbf{x}) = \begin{cases} 0 & \text{if } \mathbf{x} \in C \\ +\infty & \text{if } \mathbf{x} \notin C \end{cases}$$
   Show that (2) is equivalent to
   $$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) + \iota_C(\mathbf{x}) \tag{3}$$
2. Show that for all $\mathbf{x} \in C$, $\partial\iota_C(\mathbf{x}) = \{\mathbf{q} \in \mathbb{R}^n \ : \ \forall \mathbf{y} \in C, \langle \mathbf{q}, \mathbf{y} - \mathbf{x}\rangle \leq 0\}$ and that $\partial\iota_C(\mathbf{x})$ is a cone (it is called the *normal cone* to C at $\mathbf{x}$). Show that for all $\mathbf{x} \notin C$, $\partial\iota_C(\mathbf{x}) = \emptyset$.
3. Show that $\mathbf{x}^*$ is a solution to (3) if and only if $-\nabla f(\mathbf{x}^*) \in \partial\iota_c(\mathbf{x}^*)$.
4. Denote $\mathcal{H}_{\mathbf{w},b} = \{\mathbf{x} \in \mathcal{X} : \langle \mathbf{w}, \mathbf{x}\rangle + b = 0\}$. Compute $\partial\iota_{\mathcal{H}_{\mathbf{w},b}}(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^d$.

   (Hint: Show that we can assume $b = 0$ without loss of generality, and use the fact that $\mathcal{H}_{\mathbf{w},0}$ is a vector space.)
5. Prove that the distance of a point $\mathbf{z}$ to $\mathcal{H}$ is equal to

$$\mathrm{d}(\mathbf{z}, \mathcal{H}_{\mathbf{w},b}) = \min_{\mathbf{x}\in\mathcal{H}_{\mathbf{w},b}} \|\mathbf{x} - \mathbf{z}\|_2 = \frac{|\langle \mathbf{w}, \mathbf{z}\rangle + b|}{\|\mathbf{w}\|_2}\,.$$

2 Solutions

Exercise 1 1. It is obvious that $\mathbf{x} \mapsto \sum_{i=1}^n f_i(\mathbf{x}) + \iota_C(\mathbf{x})$ can admit a minimum only on C (the f_i are differentiable, so in particular they take finite values). On the other hand, if $\mathbf{x} \in C$, this function is the same as $\mathbf{x} \mapsto \sum_{i=1}^n f_i(\mathbf{x})$. Thus the constrained minimization problem is the same as the unconstrained one with the convex indicator function.

2. It is clear that:

$$\mathrm{prox}_{\iota_C}(\mathbf{x}) = \underset{\mathbf{y}\in\mathbb{R}^d}{\mathrm{argmin}}\Big\{\iota_C(\mathbf{y}) + \frac{1}{2}\|\mathbf{x}-\mathbf{y}\|^2\Big\} = \underset{\mathbf{y}\in C}{\mathrm{argmin}}\ \|\mathbf{x}-\mathbf{y}\|^2.$$

And it is well known that the point minimizing this distance (to the closed convex set $C = [0,1]^d$) is the projection of $\mathbf{x}$ on C :

$$\mathrm{prox}_{\iota_C}(\mathbf{x}) = \mathrm{proj}_C(\mathbf{x}).$$

3. Notice that:

$$\sum_{i=1}^n f_i(\mathbf{x}) = n\mathbb{E}[f_I(\mathbf{x})] = \mathbb{E}[nf_I(\mathbf{x})],$$

where I is a random variable with uniform distribution on $[\![1,n]\!]$. In other words, we choose at random function among the n functions f_i, we evaluate it at $\mathbf{x}$, and thus define a random function. Then (1) is equivalent to minimizing n times the expectation of this random function with respect to $\mathbf{x}$.

In this context, the ξ_k (using the notations of the lecture) will have the uniform distribution $\mathcal{U}_{[\![1,n]\!]}$. So we will apply the stochastic proximal gradient method to $(\mathbf{x},\xi_k) \mapsto nf_{\xi_k}(\mathbf{x}) + \iota_C(\mathbf{x})$. It is just the usual proximal gradient algorithm, but applied to a uniformly chosen function nf_i, with a penalization function given by[1] $g = \iota_C$. Finally we find the following algorithm:

$$\mathbf{x}_{k+1} = \mathrm{proj}_C\big(\mathbf{x}_k - \gamma_k n\nabla f_{\xi_k}(\mathbf{x}_k)\big),$$

where γ_k is a sequence of step sizes.

Exercise 2 1. It is the same line of reasoning as in the first question of the previous exercise.

2. • For each $\mathbf{x} \in \mathbb{R}^d$, we must find all vectors $\mathbf{q} \in \mathbb{R}^d$ such that:

$$\forall \mathbf{y} \in \mathbb{R}^d,\ \iota_C(\mathbf{y}) \geq \iota_C(\mathbf{x}) + \langle \mathbf{q}, \mathbf{y}-\mathbf{x}\rangle. \tag{4}$$

If $\mathbf{x} \in C$, then the only non-trivial case is when $\mathbf{y} \in C$ too. So $\mathbf{q}$ must satisfy:

$$0 \geq 0 + \langle \mathbf{q}, \mathbf{y}-\mathbf{x}\rangle = \langle \mathbf{q}, \mathbf{y}-\mathbf{x}\rangle.$$

This condition is clearly both necessary and sufficient, so:

$$\partial\iota_C(\mathbf{x}) = \{\mathbf{q} \in \mathbb{R}^d \mid \langle \mathbf{q}, \mathbf{y}-\mathbf{x}\rangle \leq 0\}.$$

Furthermore this set is indeed a cone: if $q \in \partial\iota_C(\mathbf{x})$, then it is evident that $\lambda\mathbf{q} \in C$ if $\lambda \geq 0$.

[1] Notice that g is convex (because C is convex) and l.s.c. (because C is closed). Also $\forall\gamma > 0,\ \gamma g = g$!

- And if $\mathbf{x} \notin C$, (4) is obviously impossible since $\iota_C(\mathbf{y}) = 0$ if $\mathbf{y} \in C$, while $\iota_C(\mathbf{y}) = +\infty$. Thus, for $\mathbf{x} \notin C$, $\partial \iota_C(\mathbf{x}) = \emptyset$.

3. A solution $\mathbf{x}^*$ necessarily belongs to C thanks to the first question. Thus the subgradient of ι_C at $\mathbf{x}^*$ exists. And since f is differentiable, Fermat's rule gives that:

$$\mathbf{x}^* \text{is a solution of (3)} \iff \mathbf{0}_{\mathbb{R}^d} \in \partial(f + \iota_C)(\mathbf{x}^*) = \{\nabla f(\mathbf{x}^*)\} + \{\mathbf{q} \in \mathbb{R}^d \mid \langle \mathbf{q}, \mathbf{y} - \mathbf{x} \rangle \leq 0\}.$$

Hence, there is $\mathbf{q} \in \partial \iota_C(\mathbf{x}^*)$ such that:

$$\mathbf{0}_{\mathbb{R}^d} = \nabla f(\mathbf{x}^*) + \mathbf{q},$$

and so $-\nabla f(\mathbf{x}^*) \in \partial \iota_C(\mathbf{x}^*)$.

4. For each $\mathbf{x} \in \mathcal{H}_{\mathbf{w},b}$, we want to find all $\mathbf{q} \in \mathbb{R}^d$ such that:

$$\forall \mathbf{y} \in \mathcal{H}_{\mathbf{w},b},\ \langle \mathbf{q}, \mathbf{y} - \mathbf{x} \rangle \leq 0. \tag{5}$$

Let $\mathbf{x}_0 \in \mathcal{H}_{\mathbf{w},b}$ and notice that:

$$\langle \mathbf{q}, \mathbf{y} - \mathbf{x} \rangle = \big\langle \mathbf{q}, (\mathbf{y} - \mathbf{x}_0) - (\mathbf{x}_0 - \mathbf{x}) \big\rangle.$$

Since $\mathbf{y} - \mathbf{x}_0$ and $\mathbf{x}_0 - \mathbf{x}$ belong to $\mathcal{H}_{\mathbf{w},b}$, we may assume that $\mathbf{x}$ and $\mathbf{y}$ belong to the vector space $\mathcal{H}_{\mathbf{w},0} = \mathrm{Span}(\mathbf{w})^\perp$. But then $\lambda \mathbf{y} \in \mathcal{H}_{\mathbf{w},0}$ too for all $\lambda \in \mathbb{R}$. In particular the inequality (5) reads:

$$\forall \mathbf{y} \in \mathcal{H}_{\mathbf{w},b},\ \forall \lambda \in \mathbb{R},\ \lambda \langle \mathbf{q}, \mathbf{y} \rangle \leq \langle \mathbf{q}, \mathbf{x} \rangle,$$

which is possible only if $\langle \mathbf{q}, \mathbf{y} \rangle = 0$ for all $\mathbf{y} \in \mathcal{H}_{\mathbf{w},0}$. On the other hand, it is easy to see that $\lambda \mathbf{w} \in \partial \iota_{\mathcal{H}_{\mathbf{w},b}}$ for all $\lambda \in \mathbb{R}$. As a result:

$$\partial \iota_{\mathcal{H}_{\mathbf{w},b}} = \partial \iota_{\mathcal{H}_{\mathbf{w},0}} = \mathrm{Span}(\mathbf{w})^{\perp\perp} = \mathrm{Span}(\mathbf{w}).$$

5. Here the function to minimize[2] is $f_{\mathbf{z}} : \mathbf{x} \mapsto \|\mathbf{x} - \mathbf{z}\|_2^2$, and the constraint is that $\mathbf{x}$ must belong to the convex set $\mathcal{H}_{\mathbf{w},b}$. So we can apply question 3 to find the minimizer. We must find a point $\mathbf{x}^*$ such that $-\nabla f(\mathbf{x}^*) = -2(\mathbf{x}^* - \mathbf{z}) \in \partial \iota_C(\mathbf{x}^*) = \mathrm{Span}(\mathbf{w})$. That is, for some $\lambda \in \mathbb{R}$,

$$\mathbf{x}^* = \mathbf{z} - \frac{\lambda}{2}\mathbf{w}.$$

To find λ, remember that $\mathbf{x}^*$ must belong to $C = \mathcal{H}_{\mathbf{w},b}$, so we must have:

$$-b = \langle \mathbf{w}, \mathbf{x}^* \rangle = \langle \mathbf{w}, \mathbf{z} \rangle - \frac{\lambda}{2}\|\mathbf{w}\|_2^2,$$

and so $\lambda = 2(b + \langle \mathbf{w}, \mathbf{x}^* \rangle)/\|\mathbf{w}\|_2^2$. Thus:

$$\mathbf{x}^* = \mathbf{z} - \frac{b + \langle \mathbf{w}, \mathbf{x}^* \rangle}{\|\mathbf{w}\|_2^2}\mathbf{w}.$$

Evaluating $\mathbf{x} \mapsto \|\mathbf{x} - \mathbf{z}\|$ in $\mathbf{x}^*$, we find the desired result.

[2] Notice the square! Without it, this function is not differentiable at $\mathbf{z}$...